20261020 Quiz

for (MATH301) Modern Algebra I

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Exercise 3.1.14. Consider the additive group $\mathbb{Q}/\mathbb{Z}$.

(a) Show that every coset has a unique representative $q \in \mathbb{Q}$ with $0 \leq q < 1$.

(b) Show every element has finite order, and orders are unbounded.

(c) Show $\mathbb{Q}/\mathbb{Z}$ is the torsion subgroup of $\mathbb{R}/\mathbb{Z}$.

(d) Show $\mathbb{Q}/\mathbb{Z}$ is isomorphic to the group of complex roots of unity.

Solution.

(a) For $q \in \mathbb{Q}$, subtract its floor $m = \lfloor q \rfloor$. Then $q - m \in [0, 1) \cap \mathbb{Q}$ and represents the same coset. If $u, v \in [0, 1)$ represent the same coset, then $u - v \in \mathbb{Z} \cap (-1, 1) = \{0\}$, so $u = v$.

(b) For $q = a/b$ with $b > 0$, the coset $q + \mathbb{Z}$ satisfies $b(q + \mathbb{Z}) = \mathbb{Z}$, hence has finite order. The coset $1/n + \mathbb{Z}$ has order exactly n for each positive integer n, so the orders are unbounded.

(c) An element $x + \mathbb{Z} \in \mathbb{R}/\mathbb{Z}$ has finite order precisely when $nx \in \mathbb{Z}$ for some positive integer n, equivalently $x \in \mathbb{Q}$. Hence the torsion subgroup of $\mathbb{R}/\mathbb{Z}$ is $\mathbb{Q}/\mathbb{Z}$.

(d) Define $\varphi : \mathbb{Q}/\mathbb{Z} \to \mathbb{C}^\times$ by $\varphi(q + \mathbb{Z}) = e^{2\pi i q}$. It is well defined because $e^{2\pi i(q+m)} = e^{2\pi i q}$ for $m \in \mathbb{Z}$, and it is a homomorphism because exponentials convert sums into products. Its kernel is trivial since $e^{2\pi i q} = 1$ exactly when $q \in \mathbb{Z}$. Every root of unity of order n is $e^{2\pi i k/n}$ for some integer k, so φ maps onto the group of all roots of unity. This gives the desired isomorphism.

□

Exercise 3.1.17. Let $G = D_8 = \langle r, s \mid r^8 = s^2 = 1,\ rs = sr^{-1} \rangle$, the dihedral group of order 16 (whose subgroup lattice appears in Section 2.5). The subgroup $\langle r^4 \rangle = Z(G)$ is normal. Let $\overline{G} = G/\langle r^4 \rangle$, and write $\bar{r} = r\langle r^4 \rangle$, $\bar{s} = s\langle r^4 \rangle$.

(a) Show $|\overline{G}| = 8$.

(b) List every element as $\bar{s}^a \bar{r}^b$.

(c) Give the order of every element in (b).

(d) Express $\overline{rs}$, $\overline{sr^{-2}s}$, and $\overline{s^{-1}r^{-1}sr}$ in this form.

(e) Show $\overline{H} = \langle \bar{s}, \bar{r}^2 \rangle$ is normal and isomorphic to V_4; identify its complete preimage in G.

(f) Find $Z(\overline{G})$ and identify $\overline{G}/Z(\overline{G})$.

Notation. The D_8 appearing here is the dihedral group of order 16 under the MATH000 convention. The textbook may index a dihedral group by its order, writing this group as D_{16}.

Solution. The element r^4 is central: it commutes with r and $sr^4s^{-1} = r^{-4} = r^4$. Hence $\langle r^4\rangle \trianglelefteq G$, and the quotient is defined.

(a) Since $|\langle r^4\rangle| = 2$, Lagrange's theorem gives $|\overline{G}| = 16/2 = 8$.

(b) The relation $\bar{r}^4 = 1$ and the dihedral normal form give

$$1, \bar{r}, \bar{r}^2, \bar{r}^3, \bar{s}, \bar{s}\bar{r}, \bar{s}\bar{r}^2, \bar{s}\bar{r}^3.$$

These eight elements are distinct: the first four are distinct powers of $\bar{r}$, and the last four belong to the different coset $\bar{s}\langle\bar{r}\rangle$.

(c) The orders in this order are

$$\boxed{1, 4, 2, 4, 2, 2, 2, 2}.$$

Indeed $|\bar{r}| = 4$, so $|\bar{r}^2| = 2$ and $|\bar{r}^3| = 4$. Moreover, $(\bar{s}\bar{r}^j)^2 = \bar{s}\bar{r}^j\bar{s}\bar{r}^j = \bar{r}^{-j}\bar{r}^j = 1$, and none of these reflections is the identity.

(d) Using $r^j s = sr^{-j}$ and $s^2 = 1$, we find

$$\overline{rs} = \bar{s}\bar{r}^3, \qquad \overline{sr^{-2}s} = \bar{r}^2, \qquad \overline{s^{-1}r^{-1}sr} = \bar{r}^2.$$

Indeed, $rs = sr^{-1}$ gives

$$rs = sr^{-1}, \qquad sr^{-2}s = r^2, \qquad s^{-1}r^{-1}sr = sr^{-1}sr = r^2.$$

In the quotient, $\bar{r}^{-1} = \bar{r}^3$, giving the stated normal forms.

(e) We have

$$\overline{H} = \{1, \bar{r}^2, \bar{s}, \bar{s}\bar{r}^2\}.$$

It has index 2, so it is normal. Since $\bar{s}$ and $\bar{r}^2$ are commuting involutions, its three nonidentity elements have order 2, giving $\overline{H} \cong V_4$. Its full preimage under $G \to \overline{G}$ is $\langle r^2, s\rangle$; r^2 has order 4, s has order 2, and $sr^2s^{-1} = r^{-2}$. It is therefore the dihedral group of order 8, namely D_4 under MATH000.

(f) An element $\bar{r}^j$ commutes with $\bar{s}$ precisely when $\bar{r}^{2j} = 1$, so j is even. No reflection can be central because it does not commute with $\bar{r}$ of order 4. Thus $Z(\overline{G}) = \langle\bar{r}^2\rangle$. In the quotient by this center, the commuting images of $\bar{r}$ and $\bar{s}$ are two distinct involutions; the quotient has order 4, so

$$\overline{G}/Z(\overline{G}) \cong V_4.$$

□

Exercise 3.1.22.

(a) Show that if $H, K \trianglelefteq G$, then $H \cap K \trianglelefteq G$.

(b) Show that any nonempty intersection of normal subgroups of G is normal, even when the collection is infinite.

Solution.

(a) The intersection is a subgroup. For $g \in G$,

$$g(H \cap K)g^{-1} = (gHg^{-1}) \cap (gKg^{-1}) = H \cap K,$$

so it is normal.

(b) Let $(N_i)_{i\in I}$ be a nonempty family of normal subgroups. Its intersection $N = \bigcap_{i\in I} N_i$ is a subgroup, and for every $g \in G$,

$$gNg^{-1} = \bigcap_{i\in I} gN_ig^{-1} = \bigcap_{i\in I} N_i = N.$$

Thus $N \trianglelefteq G$ regardless of the size of the index set.

□

Exercise 3.1.24. Prove that if $N \trianglelefteq G$ and $H \leq G$, then $N \cap H \trianglelefteq H$.

Solution. The intersection is a subgroup of H. For $h \in H$ and $x \in N \cap H$, normality of N gives $hxh^{-1} \in N$, while closure of H gives $hxh^{-1} \in H$. Thus $h(N \cap H)h^{-1} \subseteq N \cap H$. Repeating with h^{-1} gives the reverse inclusion, so $N \cap H \trianglelefteq H$.

□

Exercise 3.1.31. If $H \leq G$ and $N \trianglelefteq H$, show $H \leq N_G(N)$. Deduce that $N_G(N)$ is the largest subgroup of G in which N is normal.

Solution. For $h \in H$, normality of N in H says $hNh^{-1} = N$. Thus every $h \in H$ belongs to $N_G(N)$ and $H \leq N_G(N)$. Conversely, by definition N is normal in $N_G(N)$, since every element of the normalizer fixes N by conjugation. Consequently $N_G(N)$ is itself one of the eligible subgroups and contains every other one, making it the largest (and their join).

□

Exercise 3.1.33. Find all normal subgroups of D_4 and determine the isomorphism type of the corresponding quotients.

Hint. Use the subgroup lattice in Section 2.5, together with the fact that subgroups of index 2 are normal.

Solution. Write $D_4 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1} \rangle$. By Lagrange's theorem a proper nontrivial normal subgroup has order 2 or 4.

If a normal subgroup N has order 2, its unique nonidentity element is fixed by every conjugation (since an automorphism of C_2 is trivial), and therefore lies in $Z(D_4) = \{1, r^2\}$. Hence $N = \langle r^2 \rangle$. The five subgroups of order 2 therefore contribute only this one normal subgroup.

Every subgroup of order 4 has index 2 and is normal. The only elements of order 4 in D_4 are r and r^3, so a cyclic subgroup of order 4 must be $\langle r \rangle$. If a subgroup of order 4 is not cyclic, it is isomorphic to V_4 and hence contains three involutions. Since the only involutions are r^2 and the four reflections, it must contain r^2 and a reflection. Reflections differing by a factor of r^2 generate the same subgroup together with r^2; the even and odd reflections give precisely $\langle r^2, s \rangle$ and $\langle r^2, sr \rangle$. Thus these are all subgroups of order 4. Adding the trivial group and the whole group gives all normal subgroups. The quotients are

N	D_4/N
$\{1\}$	D_4
$\langle r^2 \rangle$	V_4
$\langle r \rangle$	C_2
$\langle r^2, s \rangle$	C_2
$\langle r^2, sr \rangle$	C_2
D_4	$\{1\}$

The three index-2 quotients have order 2. For the central quotient, the images of r and s are distinct commuting involutions, so it is V_4.

□

Exercise 3.1.34. Let $D_n = \langle r, s \mid r^n = s^2 = 1,\ rs = sr^{-1} \rangle$, $n \geq 3$, and let k be a positive divisor of n.

(a) Prove $\langle r^k \rangle \trianglelefteq D_n$.

(b) Show $D_n/\langle r^k \rangle$ is dihedral of order $2k$.

Solution.

(a) Conjugation by the generators preserves $\langle r^k \rangle$, because $r(r^k)r^{-1} = r^k$ and $s(r^k)s^{-1} = r^{-k}$. Thus $\langle r^k \rangle$ is normal in D_n.

(b) Let $\bar{r} = r\langle r^k \rangle$ and $\bar{s} = s\langle r^k \rangle$. These satisfy

$$\bar{r}^k = \bar{s}^2 = 1, \qquad \bar{s}\bar{r}\bar{s}^{-1} = \bar{r}^{-1}.$$

Since $k \mid n$, the subgroup $\langle r^k \rangle$ has order n/k, so the quotient has order $2k$. Its elements are the $2k$ distinct cosets $\bar{r}^j$ and $\bar{s}\bar{r}^j$ $(0 \le j < k)$, giving the dihedral presentation. For $k \ge 3$ the quotient is D_k in MATH000; for $k = 1, 2$ the degenerate cases are C_2 and V_4.

□

Exercise 3.1.40. Let $N \trianglelefteq G$ and $\bar{G} = G/N$. Show that $\bar{x}$ and $\bar{y}$ commute if and only if $[x, y] = x^{-1}y^{-1}xy \in N$.

Solution. The images commute if and only if

$$\bar{x}^{-1}\bar{y}^{-1}\bar{x}\bar{y} = 1_{G/N}.$$

The left-hand side equals $\overline{x^{-1}y^{-1}xy} = \overline{[x, y]}$, which equals the identity coset precisely when $[x, y] \in N$.

□

Exercise 3.1.41. Prove that $[G, G] = \langle [x, y] : x, y \in G \rangle$ is a normal subgroup of G and that $G/[G, G]$ is abelian.

Solution. Conjugation takes a commutator to a commutator:

$$g[x, y]g^{-1} = [gxg^{-1}, gyg^{-1}].$$

Consequently conjugation by any $g \in G$ preserves the subgroup generated by all commutators, so $[G, G] \trianglelefteq G$. Every commutator belongs to this subgroup. By Exercise 3.1.40 the images of any two elements of G therefore commute in the quotient $G/[G, G]$, which is abelian.

□

Exercise 3.2.4. Show that if $|G| = pq$ for primes p, q (not necessarily distinct), then either G is abelian or $Z(G) = \{1_G\}$.

Hint. Recall that if $G/Z(G)$ is cyclic, then G is abelian.

Solution. Suppose $Z(G)$ is nontrivial. If it equals G, the group is abelian. Otherwise its order is a proper divisor of pq larger than 1, necessarily p or q, so the quotient $G/Z(G)$ has prime order and is cyclic. If $G/Z(G) = \langle xZ(G) \rangle$, every element of G can be written $x^a z$ with $z \in Z(G)$. For any a, b and central z, w, the elements $x^a z$ and $x^b w$ commute. Hence G is abelian. Contrapositively, if G is not abelian its center is trivial.

□

Exercise 3.2.5. For $H \le G$ and $g \in G$:

(a) Show gHg^{-1} is a subgroup of the same order as H.

(b) If H is the unique subgroup of order n, show $H \trianglelefteq G$.

Solution.

(a) Conjugation $c_g(x) = gxg^{-1}$ is an automorphism of G. Its restriction to H is a bijection onto gHg^{-1}, and the image is a subgroup. Hence $|gHg^{-1}| = |H|$ even if H is infinite.

(b) For each $g \in G$, gHg^{-1} is a subgroup of order n. Uniqueness forces $gHg^{-1} = H$. This is exactly the definition of $H \trianglelefteq G$.

□

Exercise 3.2.8. If $H, K \leq G$ are finite subgroups of relatively prime orders, prove $H \cap K = \{1_G\}$.

Solution. The group $H \cap K$ is a subgroup of both H and K. By Lagrange's theorem, its order divides both $|H|$ and $|K|$. Since these numbers are coprime, $|H \cap K| = 1$, so $H \cap K = \{1_G\}$.

□

Exercise 3.2.11. For $H \leq K \leq G$, prove $[G : H] = [G : K][K : H]$, allowing infinite groups and infinite indices.

Solution. Choose representatives g_i $(i \in I)$ for the left K-cosets in G, and representatives k_j $(j \in J)$ for the left H-cosets in K. Every $g \in G$ belongs to exactly one coset g_iK; within it $g_i^{-1}g$ belongs to exactly one coset k_jH. Thus the cosets g_ik_jH cover G. If $g_ik_jH = g_{i'}k_{j'}H$, passing to K-cosets gives $i = i'$, and cancellation then gives $j = j'$. Consequently the left H-cosets in G are in bijection with $I \times J$, so

$$[G : H] = |I \times J| = |I|\,|J| = [G : K][K : H].$$

This is a cardinal equality, also valid for infinite indices.

□

Exercise 3.2.14. Prove that S_4 has no normal subgroup of order 8 and no normal subgroup of order 3.

Solution. Suppose $N \trianglelefteq S_4$ has order 8. Then S_4/N has order 3. Every transposition has order 2, so its image in this quotient has order dividing both 2 and 3, hence is the identity. Since transpositions generate S_4, the quotient map would be trivial, contradicting the quotient's order 3.

If $N \trianglelefteq S_4$ has order 3, it contains a 3-cycle. All eight 3-cycles in S_4 are conjugate, so normality forces N to contain all eight. Adding the identity gives at least nine elements, contradicting $|N| = 3$.

□

Exercise 3.2.16. Use Lagrange's theorem in $(\mathbb{Z}/p\mathbb{Z})^\times$ to prove Fermat's little theorem: $a^p \equiv a \pmod{p}$ for every integer a, where p is prime.

Solution. If $p \mid a$, both sides are zero modulo p. Otherwise $[a]_p \in (\mathbb{Z}/p\mathbb{Z})^\times$, a group of order $p - 1$. Lagrange's theorem gives $[a]_p^{p-1} = [1]_p$. Thus $a^{p-1} \equiv 1 \pmod{p}$. Multiplying by a yields $a^p \equiv a \pmod{p}$, as required.

□

Exercise 3.2.20. If A is an abelian normal subgroup of G and B is any subgroup, prove $A \cap B \trianglelefteq AB$.

Solution. Because $A \trianglelefteq G$, the product AB is a subgroup of G. Put $N = A \cap B$. If $b \in B$, then $bNb^{-1} \subseteq A$ by normality of A and $bNb^{-1} \subseteq B$ by closure of B, so B normalizes N (equality using b^{-1}). Every $a \in A$ centralizes N because A is abelian. Thus both A and B normalize N, hence their product AB does. Since $N \leq AB$, it follows that $N \trianglelefteq AB$.

□

Exercise 3.2.21. Prove that $(\mathbb{Q}, +)$ has no proper subgroups of finite index; deduce the same for $\mathbb{Q}/\mathbb{Z}$.

Hint. The additive group $\mathbb{Q}$ is divisible: for any $x \in \mathbb{Q}$ and positive integer m, there is y with $my = x$.

Solution. Let $H \leq \mathbb{Q}$ and $[\mathbb{Q} : H] = m < \infty$. As $\mathbb{Q}$ is abelian, $\mathbb{Q}/H$ is a group of order m, and Lagrange's theorem gives $mq \in H$ for every $q \in \mathbb{Q}$. Given any $x \in \mathbb{Q}$, apply this to $q = x/m$ to obtain $x \in H$. Thus $H = \mathbb{Q}$.

If $K \leq \mathbb{Q}/\mathbb{Z}$ has finite index, its inverse image under the quotient map $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ is a subgroup of $\mathbb{Q}$ with the same finite index. It must equal $\mathbb{Q}$, so $K = \mathbb{Q}/\mathbb{Z}$.

□

Exercise 3.3.3. If $H \trianglelefteq G$ has prime index p, prove that for every subgroup $K \leq G$ either $K \leq H$, or $G = HK$ and $[K : K \cap H] = p$.

Solution. Because H is normal, HK is a subgroup containing H. The index formula gives

$$[G : H] = [G : HK][HK : H] = p.$$

As p is prime, either $HK = H$, so $K \leq H$, or $HK = G$. In the latter case, the Second Isomorphism Theorem gives

$$K/(K \cap H) \cong HK/H = G/H,$$

whence $[K : K \cap H] = |G/H| = p$.

□

Exercise 3.3.4. Suppose $C \trianglelefteq A$ and $D \trianglelefteq B$. Prove $C \times D \trianglelefteq A \times B$ and $(A \times B)/(C \times D) \cong (A/C) \times (B/D)$.

Solution. Conjugation acts coordinatewise, so

$$(a, b)(C \times D)(a, b)^{-1} = (aCa^{-1}) \times (bDb^{-1}) = C \times D.$$

Thus the product subgroup is normal. Define $\varphi : A \times B \to (A/C) \times (B/D)$ by $\varphi(a, b) = (aC, bD)$. The map is a surjective homomorphism whose kernel is $C \times D$, and the First Isomorphism Theorem yields the required isomorphism.

□

Exercise 3.5.3. Prove that S_n is generated by the adjacent transpositions $s_i = (i\ i + 1)$, $1 \leq i < n$.

Hint. A conjugate of a transposition is another transposition: $\sigma(a\ b)\sigma^{-1} = (\sigma(a)\ \sigma(b))$.

Solution. Every permutation is a product of transpositions, so it is enough to express an arbitrary transposition through adjacent ones. If $1 \leq i < j \leq n$, then

$$(i\ j) = s_i s_{i+1} \cdots s_{j-2} s_{j-1} s_{j-2} \cdots s_{i+1} s_i.$$

To see this, put $w = s_i \cdots s_{j-2}$. Then w fixes j and sends $j-1$ to i (under right-to-left composition), so $ws_{j-1}w^{-1}$ interchanges i and j. The displayed word is exactly $ws_{j-1}w^{-1}$. Thus every transposition, and hence every permutation, is generated by the adjacent transpositions. □

Exercise 3.5.4. Show that $S_n = \langle (12), (1\ 2\ \cdots\ n) \rangle$ for all $n \geq 2$.

Solution. Put $s = (12)$ and $c = (1\ 2\ \cdots\ n)$. For $1 \leq i < n$,

$$c^{i-1} s c^{-(i-1)} = (i\ i+1).$$

Thus $\langle s, c \rangle$ contains every adjacent transposition. By Exercise 3.5.3, these generate the whole S_n, proving the claim.

□

Exercise 5.1.1. Prove $Z(G_1 \times \cdots \times G_n) = Z(G_1) \times \cdots \times Z(G_n)$. Deduce the product is abelian if and only if every factor is abelian.

Solution. Multiplication in a direct product is coordinatewise. An element $(g_1, \ldots, g_n)$ commutes with every $(h_1, \ldots, h_n)$ precisely when $g_i h_i = h_i g_i$ for every i and every $h_i \in G_i$. This means $g_i \in Z(G_i)$ for each coordinate, proving the formula. The product is abelian exactly when its center is the whole product, which by this formula happens exactly when $Z(G_i) = G_i$ for all i. □

Exercise 5.1.14. For $G = A_1 \times \cdots \times A_n$ and $B_i \trianglelefteq A_i$, prove $B_1 \times \cdots \times B_n \trianglelefteq G$ and

$$G/(B_1 \times \cdots \times B_n) \cong (A_1/B_1) \times \cdots \times (A_n/B_n).$$

Solution. Conjugation in the direct product is coordinatewise. Thus

$$(a_i)(\prod_{i=1}^{n} B_i)(a_i)^{-1} = \prod_{i=1}^{n} a_i B_i a_i^{-1} = \prod_{i=1}^{n} B_i,$$

so $\prod_i B_i$ is normal. The map

$$\varphi : G \to \prod_{i=1}^{n} (A_i/B_i), \qquad (a_i) \mapsto (a_i B_i)$$

is a surjective homomorphism: any cosets have representatives in their respective factors. Its kernel is precisely $\prod_i B_i$. The First Isomorphism Theorem proves the result.

□

Exercise 5.2.2. For each of the following orders, give the invariant-factor lists for all abelian groups of that order:

(a) 270;

(b) 9801;

(c) 320;

(d) 105;

(e) 44100.

Solution. We use decreasing invariant factors: $(n_1, \dots, n_t)$ denotes $C_{n_1} \times \cdots \times C_{n_t}$ with $n_{i+1} \mid n_i$. The classification theorem says to partition the exponents in the prime factorization, combine the corresponding prime powers in aligned columns, and omit factors equal to 1. We obtain:

(a) $270 = 2 \cdot 3^3 \cdot 5$. Since 3 has partitions 3, $2+1$, and $1+1+1$, there are three types:

$$(270), \qquad (90, 3), \qquad (30, 3, 3).$$

(b) $9801 = 3^4 \cdot 11^2$. There are $p(4)p(2) = 5 \cdot 2 = 10$ types:

$$\begin{array}{ccc} (9801) & (3267, 3) & (1089, 9) \\ (1089, 3, 3) & (891, 11) & (363, 3, 3, 3) \\ (297, 33) & (99, 99) & (99, 33, 3) \\ & (33, 33, 3, 3). & \end{array}$$

(c) $320 = 2^6 \cdot 5$. The exponent 6 has $p(6) = 11$ partitions, hence eleven types:

$$\begin{array}{ccc} (320) & (160, 2) & (80, 4) \\ (80, 2, 2) & (40, 8) & (40, 4, 2) \\ (40, 2, 2, 2) & (20, 4, 4) & (20, 4, 2, 2) \\ (20, 2, 2, 2, 2) & (10, 2, 2, 2, 2, 2). & \end{array}$$

(d) $105 = 3 \cdot 5 \cdot 7$ has only one type because every exponent is 1:

$$(105).$$

(e) $44100 = 2^2 \cdot 3^2 \cdot 5^2 \cdot 7^2$. Each exponent 2 has two partitions, so there are $2^4 = 16$ types:

$$\begin{array}{ccc} (44100) & (22050, 2) & (14700, 3) \\ (8820, 5) & (7350, 6) & (6300, 7) \\ (4410, 10) & (3150, 14) & (2940, 15) \\ (2100, 21) & (1470, 30) & (1260, 35) \\ (1050, 42) & (630, 70) & (420, 105) \\ & (210, 210). & \end{array}$$

Every displayed list has product equal to the prescribed order and successive factors dividing preceding ones. The classification theorem gives uniqueness of the invariant-factor type; the partition counts $3, 10, 11, 1, 16$ match the displayed lists, so none is missing.

□

Exercise 5.2.5. If a finite abelian group G has invariant factors $(n_1, n_2, \dots, n_r)$ with $n_{i+1} \mid n_i$, prove G contains an element of order m if and only if $m \mid n_1$. Deduce that the exponent of G is n_1.

Solution. Write $G \cong C_{n_1} \times \cdots \times C_{n_r}$. The order of an element is the least common multiple of the orders of its coordinates. Since the order of the ith coordinate divides n_i and $n_i \mid n_1$, every element has order dividing n_1. Conversely, whenever $m \mid n_1$, the first cyclic factor contains an element of order m; adjoining identity elements in the remaining coordinates gives an element of order m in G. In particular, the first factor supplies an element of order n_1, and every element order divides n_1. Hence the exponent (least common multiple of all element orders) is n_1.

□

Exercise 5.2.9. Let $A = C_{60} \times C_{45} \times C_{12} \times C_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in A.

Solution. In a cyclic group of even order, the equation $x^2 = 1$ has exactly two solutions; in an odd-order cyclic group it has only one. Exactly three factors of A have even order. Hence $2^3 = 8$ elements satisfy $x^2 = 1$, and excluding the identity leaves $\boxed{7}$ elements of order exactly 2.

Every subgroup of index 2 is normal and is the kernel of a surjective homomorphism $A \to C_2$. Such a homomorphism is determined by its values on the four cyclic generators: each even-order factor has two choices for the image, while the odd-order factor has only the identity choice. Hence there are $2^3 - 1 = 7$ nontrivial homomorphisms, all surjective. Two surjections $A \to C_2$ with the same kernel agree, since their induced maps $A/K \to C_2$ differ by an automorphism of C_2, and that group has only the identity automorphism. Thus there are $\boxed{7}$ distinct index-2 subgroups.

□